

Figure 1 is a line plot showing the dispersion energy (in kcal/mol) of the (H₂O)₆ cluster as a function of distance (in Å). The y-axis ranges from 0 to -10 kcal/mol, and the x-axis ranges from 1.5 to 4.0 Å. A vertical dashed line is drawn at 1.9 Å, indicating the experimental distance. The plot compares six different computational methods:

- SAPT(PBE0)-D4 (S) (Red line with square markers)
- SAPT(PBE0)-D4 (I) (Green line with triangle markers)
- HF-D4 (Blue line with circle markers)
- SAPT(PBE0)/aDZ (Brown line with cross markers)
- SAPT(PBE0)/aTZ (Grey line with plus markers)
- SAPT2+3(CCD)/aTZ (Black line with circle markers)

The curves show that the dispersion energy decreases (becomes more negative) as the distance decreases, reaching a minimum around 1.9 Å. The HF-D4 method (blue line) consistently yields the highest dispersion energy, while the SAPT2+3(CCD)/aTZ method (black line) yields the lowest dispersion energy at the minimum distance.